*Notes in*

# Pediatrics

## History Taking

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https://goo.gl/RpvNsl



# Subject1: History questions

## Identification:

- Name of baby (triple name).
- Age and datell of birth.
- Gender.
- Blood group
- Source of history (mother – father – grandmother).
- Occupation of parents.
- Residence of parents.
- Religion of parents.

## Date of admission.

## Date of examination.

## Chief compliant:

- Mention the main reason that brings the patient to the hospital (up to 2 chief compliant).
- Duration of chief compliant.

## History of present illness:

- Last time the patient was **well.**
- The **story** (take it from the source like the mother, ask her to tell you the full story then summarize it and write it in her words).
- **Analysis** of the symptoms.
- Complete the **same system**.
- **Relevant systems** (ask about any system related to the involved system).
- **Routine questions**: always ask about (fever, sweating, urine output, feeding, activity, sleep, weight loss).
- Ask about attention to **private doctor or primary health centers PHC** (ask about investigations, treatment, improvement or not).
- **Admission** (ask about investigations, treatment, improvement or not).
- Any **new event** that may occur during hospitalization ((worsening of baby condition ask about sleep + activity (smile/fatigue) + feeding all will decreased)).
- **Outcome** (the condition of patient at the moment of taking history).

((Convert these points to story and write it))

**Systems review:** (questions here depend on the age of baby)

- **GIT** → abdominal pain, diarrhea, constipation, vomiting, hematemesis, melena, jaundice, nausea and loss of appetite (the last two describe as poor feeding).
- **Respiratory** → SOB, noisy breathing, cough, hemoptysis, sputum.
- **CVS** → dyspnea, palpitation (the mother could feel palpitation of her baby), cyanosis.
- **Genitourinary** → color and amount of urine, hematuria, pyuria, dysuria (crying during urination), frequency, any abnormality in the genetalia.
- **Nervous system** → headache, convulsions, abnormal movements, abnormal hearing or vision.
- **Loco-motor** → joint pain and stiffness, joint swelling, abnormal movements, restricted movements.
- **Hematology** → epistaxis, bruises, patichae, bleeding from anywhere.
- **Skin** → dryness, discoloration, pigments, itching, rash, lump, hair and nail changes.

## Pre-natal (Ante-natal):

- ANC (attend regular visits to PHC, take folic acid and supplements).
- Disease of mother during pregnancy (infections like Toxoplasma-HIV-Rubella-Cytomegalovirus-Chickenpox-Hepatitis --- TORCH infection appear as fever + skin rash + joint swelling --- other diseases like D.M and hypertension and anemia).
- Drugs taken during pregnancy.
- Exposure to radiation during pregnancy.
- Smoking.
- Bleeding.
- Previous abortion or neonatal death.
- Admission to hospital for any reason.
- Vaccine.
- Number of child.

## Natal:

- Place of delivery (at home or hospital).
- Type of delivery (vaginal or caesarian section CS) if CS what is the indication? Elective or emergency?
- Duration of delivery (normal or prolonged).
- Any complication during labor.
- Instrument use in the delivery.
- Gestational age (pre-term or term or post-date or post-term).
- Weight of the baby (normal AGA, Small SGA, Large LGA).

**Post-natal:** (first 28 days of life)

- Immediate crying.

- Time of discharge from the hospital (24 hours – 48 – more).
- NICU admission.
- When the baby passed urine (within first 24 hours) and meconium (the first stool) (within first 48 hours).
- Movement.
- Neonatal jaundice, Cyanosis (fetal distress), Fit, bleeding.
- Baby developed other diseases? Treatment?

## Past-medical history:

- Previous similar disease.
- Previous admissions.
- Chronic diseases.
- Previous infections (measles, others).
- Blood transfusion.

## Past-surgical:

- Previous operations (indication, type, outcome).
- Circumcision in male and ear piercing in female (at which age, any complications)…
- Hospitalization.

**Complications of circumcision:**

- Bleeding.
- Infection.
- Insufficient Foreskin Removed.
- Excessive Foreskin Removed.
- Adhesions / Skin Bridges.
- Inclusion Cysts.
- Abnormal Healing.
- Meatitis.
- Meatal Stenosis.
- Urinary Retention.
- Phimosis.
- Chordee.
- Hypospadias.
- Epispadias.
- Urethrocutaneous Fistula.
- Necrosis of the Penis.
- Amputation of the Glans.
- Death.

Source: med.stanford.edu

## Drug history:

- Drug taking by mother (chronic use).
- Drug taking by baby (chronic use).
- Allergy to drug and other substances.

## Feeding history:

1- Breast feeding:

- Way of feeding (using both right and left breast each feeding time).
- Regular (at least every 3 hours) or on demand.
- Any problem with feeding (large nipple, others).

2- Bottle feeding:

- Way of feeding.
- Way of preparation.
- Type of formula use (lactose free, soy milk formula, others).
- Way of sterilization of the bottle (boiling, Washing, brushing).
- Number of bottles.
- Number of feeding.
- Regular (at least every 3 hours) or on demand.

- Any problem occur after bottle feeding (diarrhea, others).
- Put the bottle in freeze for cooling.

3- Mixed feeding (breast and bottle feeding).

4- Semi-solid or solid food:
   - At which age given.
   - Type.
   - Any problem occur after this feeding.

5- Weaning: at which age milk was taken off his diet.

6- Pica: Ask if the child eat soil, wood or other things (caused by iron deficiency anemia – Ca deficiency – lead poisoning).

## Immunization history:

- Take the vaccines on regular Iraqi schedule or not?
- On national immunization day only?
- Type of vaccine? , At which age?
- Any complications (fever, pain, convulsion, rash, excessive crying) pertussis vaccine cause (convulsion, shock).
- Time of last vaccine.
- Notes: RV not give to baby after 3 and 8 months of age / BCG not give to baby after 1 year of age / BCG should return if the scar not appear.

| **Age** | **VACCINE** |
|---|---|
| 24 hours after birth | BCG, HBV , OPV |
| 2 months | DTP-HepB-Hib, OPV, RV |
| 4 months | DTP-Hib, OPV, RV |
| 6 months | DTP-HepB-Hib, OPV, RV |
| 9 months | Measles, Vit.A 100,000 U |
| 15 months | MMR |
| 18 months | DTP-Hib, OPV, Vit.A 200,000 U |
| 4-6 years | DTP, MMR, OPV |

| **Name** | **In Arabic** | **Type** | **Route** |
|---|---|---|---|
| BCG | بي سي جي | Live attenuated TB bacteria | ID |
| HBV | إالتهاب الكبد الفيروسي ب | Inactivated hepatitis B virus | IM |
| OPV | شلل الأطفال فموي | Live attenuated Polio virus | Oral |
| RV | الفايروس العجلي | Live attenuated Rotavirus | Oral |
| Measles | الحصبة المنفردة | Live attenuated measles virus | SC |
| MMR | الحصبة المختلطة | Live attenuated, mumps, rubella viruses | SC |
| DTP | اللقاح الثلاثي | Diphtheria and Tetanus Toxoid + inactivated pertussis bacteria | IM |
| DTP-Hib | اللقاح الرباعي | DTP + Inactivated Influenza virus | IM |
| DTP-HepB-Hib | اللقاح الخماسي | DTP + Inactivated Influenza virus + Inactivated hepatitis B virus | IM |

## Developmental:

- Gross motor (sit, stand, walk) or crawl.
- Fine motor (Grasp, move object from hand to hand, etc.).
- Social (smile, laugh, etc.).
- Speech, Vision, Hearing.
- Urination and defecation control.

## Family history:

- Health of parents, brothers and sisters.
- Family history of same illness.
- Family history of chronic disease or infectious disease.
- Family history of unexplained death.
- Order of baby in the family.
- Any illness in the family.

## Social history:

- Rural or urban.
- Source of water.
- Income.
- Level of education and occupation of parents.
- Crowding index ➔ (number of persons/number of rooms) ➔ below 3 is normal and above 3 is crowding.
- Any domestic animals.
- Ventilation.
- Smoking.
- Sewage disposal.

## Obstetrical and gynecological history:

- For girls because pediatric age up to 18 years.
- 1. Age of menarche.
- 2. Any abnormality in the cycle.
- 3. Use any contraceptive or traditional methods.
- 4. Any gynecological infection or disease.
- 5. Any gynecological procedure or surgery.

# Subject2: GIT history

## 1-History of bowel motion (diarrhea):

- Amount:
  - Increased (watery or loose) in small intestine disease and infections like V.Cholera.
  - Decreased (small and bulky) in large intestine disease like Colitis.
- Frequency:
  - High frequency in small intestine diseases.
  - Low frequency in large intestine diseases.
- Color:
  - Normal color either yellow, brown or green.
  - Green color ➔ normal, from breastfeeding, drugs, food (Spinach or other leafy vegetables).
  - Dark green color ➔ from bile.
  - Black color ➔ stomach bleeding, food (grape juice), drugs (iron).
  - Red color ➔ lower GIT bleeding, drugs (Amoxicillin), food (beets).
  - White color ➔ biliary obstruction (no bile), milk only diet, drugs (antacids).
- Presence blood:
  - Intestinal infection.
  - Infant colitis.
  - Intussusception.
  - Inflammatory bowel disease.
  - Antibiotic-associated colitis
- Presence mucus:
  - Intussusception.
  - Inflammatory Bowel Disease.
  - Meckel's Diverticulum.
- Consistency:
  - Watery occur due to V.Cholera.
  - Loose occur in small bowel diseases.
  - Firm occur in large bowel diseases.
- Odor:
  - Not important.
  - Fish odor ➔ small intestine disease, infections (V.Cholera and viral).
  - Offensive odor ➔ Bacteria, Amebiasis, coeliac disease, lactose intolerance.
- Day or night:
  - Only at night ➔ UTI or Typhoid.
  - Only day ➔ related to feeding.
  - Day and night ➔not related to feeding and it is secretory type related to infection like Typhoid, UTI, TB, Brucellosis.
- Associated features:
  - Colic – fever – sweating – anorexia.

- Presence of Tenesmus:
  - Occur during or after defecation.
  - Causes: Constipation, Hemorrhoids, Crohn's disease, Anal abscess, Rectal abscess.
- Diarrhea with vomiting:
  - Viral infection.
  - Food poisoning.
  - Bacterial infection.
  - Traveler's diarrhea.
- Diarrhea with cough:
  - Infections: Rotavirus, adenovirus.
  - Respiratory causes: cystic fibrosis, TB.

***Notes:***

- Viral infection (Rota virus) ➔ flu like illness then vomiting then diarrhea (high amount + loose + high frequency more than 10) + low grade fever or no fever.
- V.Cholera ➔ large amount + watery + loose + high frequency.
- Shigellosis ➔ Bloody diarrhea + Tenesmus.
- Fissure in ano ➔ Fresh + large amount of blood per-rectum.
- Hypokalemia ➔ Diarrhea + abdominal distention + weakness.
- Toddler diarrhea ➔ due to excessive sweets & juices consumption, it relived by fasting.
- Parenteral diarrhea ➔ due to systemic cause rather than GIT like meningitis, UTI, Otitis
  *How to know parenteral = normal general stool + stool culture –ve + source of infection.*
- No-infectious cause of diarrhea ➔ diverticulum, ischemic colitis, hemorrhoids.
- Metabolic acidosis occur in diarrhea due to loss of HCO3- cause deep and decrease breathing to waste CO2.
- Secretory diarrhea not affected by feeding – osmotic diarrhea affected by feeding.

| **Amebic dysentery** | **Bacillary dysentery** |
|---|---|
| Caused by E.Histolytica | Caused by salmonella or shigella |
| Trophozoite in feces | Bacteria in feces |
| Gradual slow onset | Acute onset |
| Not ill (toxic) | Looks ill (toxic) |
| Long incubation period (weeks) | Short incubation period (days) |
| Pink or chocolate color | Green color |
| Abdominal pain | Abdominal pain |
| Features of large intestinal diarrhea | Features of small intestinal diarrhea |
| No systemic manifestations | There are systemic manifestations |
| **Sometimes amebic dysentery has acute form like bacterial dysentery ➔ in low immunity and old people** | |

## 2-Constipation:

- Onset and duration of decreased bowel motion.
- Frequency of defecation.
- Associated with pain, blood, mass.
- Fluid intake, type of nutrition.
- Chronic disease (like coeliac).
- Psychological problem.
- Any treatment or intervention.

***Notes:***

- Frequency of defecation:
  - Infants in first week ➔ four stools per day.
  - 1 year of age ➔ two stools per day.
  - Breast-fed infants ➔ may not pass stools for several days and be entirely healthy.
  - 4 years of age ➔ varies from three stools per day to three stools per week (similar to adults).
- Definition of constipation ➔ is the infrequent passage of dry, hardened feces often accompanied by straining or pain, there may be abdominal pain which waxes and wanes with passage of stool or overflow soiling (Emotions and behavior).
- Causes of constipation:
  - Multifactorial.
  - Arising in acutely ill young children.
  - Precipitated by dehydration or reduced fluid intake.
  - Problems with toilet training, unpleasant toilets or stress.
  - Hirschsprung disease.
  - Anorectal abnormalities.
  - Anal fissure causing pain.
  - Hypothyroidism.
  - Hypercalcemia.
- Treatment:
  - Disimpaction regimen of stool softeners for 1-2 weeks➔ initially with a macrogol laxative (polyethylene glycol) + electrolytes (Movicol Paediatric Plain).
  - If this proves unsuccessful, start stimulant laxative ➔ senna, or sodium picosulphate.
  - If the polyethylene glycol + electrolytes is not tolerated use ➔ an osmotic laxative
  - Maintenance treatment.
  - Dietary interventions ➔ sufficient fluid and a balanced diet including adequate fibre.
  - The child should be encouraged to sit on the toilet after mealtimes.
  - If fecal retention is so severe that evacuation use ➔ enemas or by manual evacuation under an anesthetic.

## 3-Vomiting:

- Onset, duration, frequency.
- Amount.
- Color (white = milk, yellow = gastric juice, green = bile).
- Consistency, Content.
- Day/night.
- Projectile or not.
- Aggravated factors (food - others).
- Associated with fever, fit, others.

***Other features:***

- Post-tussive emesis = vomiting after cough → occur in pertussis infection and mycoplasma pneumonia.
- Bile-stained vomit ➔ intestinal obstruction.
- Projectile vomiting, in first few weeks of life ➔ pyloric stenosis.
- Bacterial → diarrhea then vomiting // viral → vomiting then diarrhea.
- Vomiting after eating by 6 hours + fever at night → staph infection.
- Vomiting after eating by more than 12 hours → E.coli and it's group.
- Vomiting after eating by more than 24 hours → Salmonella and it's group.

***Notes:***

- Definitions:
  - Posseting ➔ non-forceful return of small amounts of milk which often accompany the return of swallowed air (wind).
  - Regurgitation ➔ non-forceful return of milk (larger, more frequent than posseting).
  - Vomiting ➔ forceful ejection of gastric contents.
- Causes of vomiting:
  - Gastroenteritis.
  - Feeding problems.
  - Ingestion of substance or drug.
  - Infection ➔ GIT, renal, respiratory, CNS.
  - GIT ➔ congenital anomalies, gastro-esophageal reflux, appendicitis, peptic ulcer, intestinal obstruction, coeliac disease.
  - Renal failure.
  - Raised intracranial pressure.
  - Diabetic ketoacidosis.
  - Torsion of the testis.
- Treatment:
  - Rehydration.
  - Antiemetic drugs can be used cautiously in children > 2 yr.
  - Useful drugs include (Promethazine, Metoclopramide, Ondansetron).
  - Treat the underlying cause.

## 4-Fever:

- Onset, duration, frequency:
  - Continuous fever (typhoid).
  - Intermittent fever (malaria).
  - Remittent (infective endocarditis).
  - Pel-ebstein fever (Hodgkin's lymphoma).
- Degree (documented or not).
- Time:
  - During night like TB which associated with sweating or during day.
- Associated with sweating, shivering, rigor.
- Reliving factors: spontaneously or by antipyretic agents (paracetamol or ibuprofen).
- Other points:
  - Ask about febrile convulsion and otitis media.
  - Presence of rash.
  - Illness of other family members.
  - Specific illness is prevalent in the community.
  - Recent travel abroad (malaria, typhoid).
  - Contact with animals (brucellosis).
  - Increased susceptibility from immunodeficiency.

## 5-Jaundice:

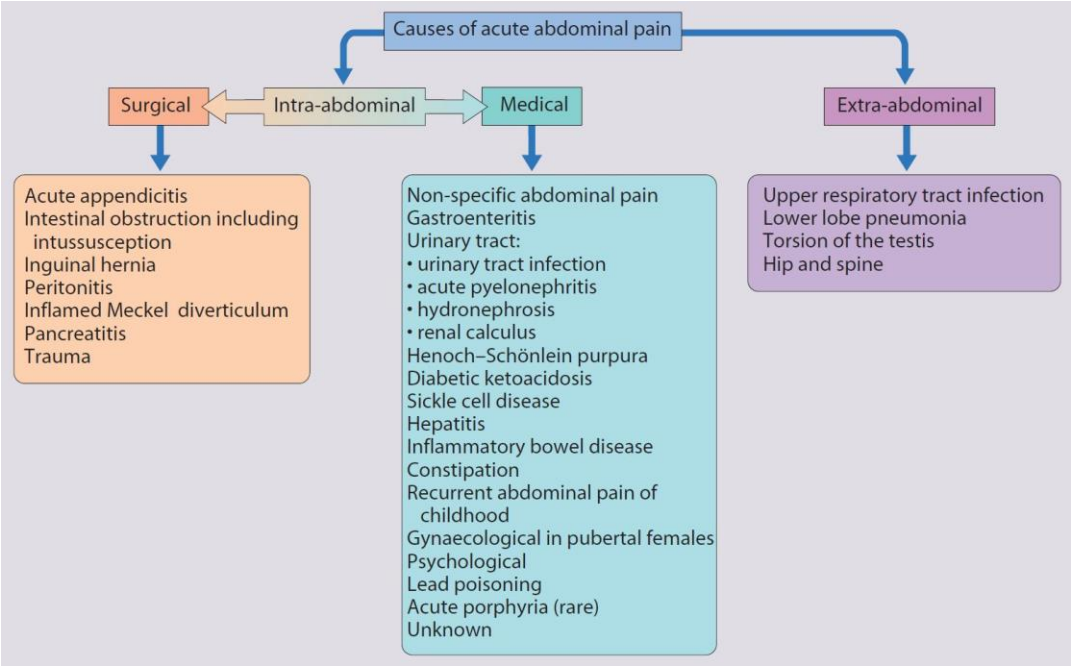
- Onset and duration (in first day or not?).
- Color of skin, urine, stool.
- Feeding problem.
- Associated with fever, vomiting, abdominal pain or mass.
- Itching (in obstructive jaundice).
- Blood group of child and parents.
- Recent blood transfusion.
- Any medications used.
- Recent disease or infection.
- Family history of liver disease.
- Mother infection during pregnancy or labor.

***Causes of jaundice:***

- Physiological jaundice.
- Breastfeeding.
- Internal bleeding (hemorrhage).
- Blood incompatibility.
- Infection.
- Liver malfunction or enzyme deficiency.
- Abnormality of baby's red blood cells.
- Others (hypothyroidism, UTI, blockage or problem in the bile ducts and gallbladder).

## 6-Abdominal pain:

- **S**ite.
- **O**nset (sudden or gradual, progressive or regressive).
- **C**haracter (ache, stabbing).
- **R**adiation.
- **A**ssociations (any other signs or symptoms).
- **T**ime course.
- **E**xacerbating/Relieving factors.
- **S**everity.




**Causes and assessment of the child with recurrent abdominal pain**

**>90% no structural cause identified**

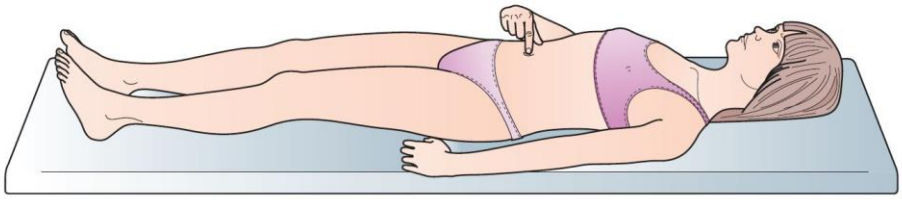
**Gastrointestinal**
- Irritable bowel syndrome
- Constipation
- Non-ulcer dyspepsia
- Abdominal migraine
- Gastritis and peptic ulceration
- Inflammatory bowel disease
- Malrotation

**Gynaecological**
- Dysmenorrhoea
- Ovarian cysts
- Pelvic inflammatory disease

**Psychosocial** – bullying, abuse, stress, etc. – a small proportion

**Hepatobility/pancreatic**
- Hepatitis
- Gall stones
- Pancreatitis

**Urinary tract**
- Urinary tract infection
- Pelvi-ureteric junction (PUJ) obstruction



**Symptoms and signs that suggest organic disease:**
- Epigastric pain at night, haematemesis (duodenal ulcer)
- Diarrhoea, weight loss, growth failure, blood in stools (inflammatory bowel disease)
- Vomiting (pancreatitis)
- Jaundice (liver disease)
- Dysuria, secondary enuresis (urinary tract infection)
- Bilious vomiting and abdominal distension (malrotation)

# Subject3: Renal history

## 1-Amount of urine:

- Anuria is absence of any urine output.
- Oliguria is decreased urine output.
  - Infants: < 0.5 mL/kg per hour for 24 hours.
  - Older children: < 500 mL/1.73 m2 body surface area per day.
- Polyuria is defined as urine output > 2000ml per 1.73M2 /24 hr or more than 2.5-3 ml/kg/24 hrs.
- Common causes of polyuria:
  - Polydipsia.
  - Solute diuresis e.g. glucosuria in diabetes mellitus.
  - Vasopressin or ADH deficiency.
  - Chronic renal failure with concentration defect.
  - Structural malformations of kidney.

## 2-Color of urine:

- Red:
  - Hematuria.
  - Myoglobinuria ➔ associated with ischemic muscle damage, crush injuries, and after vigorous exercise.
  - Hemoglobinuria ➔ with intravascular hemolysis.
  - Foods ➔ Berries, Beets, Food coloring.
  - Drugs ➔ Dilantin, Phenothiazines.
- Orange:
  - Pyridium, Rifampin.
  - Urate crystals.
- Green:
  - Food colorings.
  - Increased carotene in the diet.
- Brown or Black
  - Hemoglobin, Bilirubin.
  - Flagyl, Nitrofurans.
  - Alkaptonuria (on standing, the urine will turn a dark color).
- Yellow:
  - Bile pigments.
- Blue:
  - Methylene blue.
- Milky or Cloudy Color:
  - Pus cells and bacteria in the urine.
  - After sitting, urine may form crystals that makes urine appear cloudy.

## 3-Pain:

- Painful micturition:
  - Urinary tract infection.
  - Candida infection.
  - Pinworms.
  - Irritative dermatitis.
  - Cystitis.
  - Urethritis.
  - Psychogenic.
- Flank pain:
  - Pyelonephritis.
  - Injury.
- Renal colic:
  - Stone.

## 4-Voiding symptoms:

- Include ➔ poor stream, hesitancy, intermittent flow and straining when passing urine.
- Causes ➔ Cystitis, Stone in urethra, Injury, Stricture, Adhesion.

## 5-Storage symptoms:

- Include ➔ increased frequency and urgency of passing urine, urge incontinence and needing to get up to pass urine at night (nocturnal enuresis).
- Causes of frequency:
  - Anxiety.
  - Drinking caffeinated beverages or fizzy drinks.
  - Emotional upset.
  - Having problems with constipation.
  - Frequent urinary tract infections.
  - Nerve damage or malfunction that causes a child to have difficulty recognizing a full bladder refraining from completely emptying the bladder when on the toilet.
- Organic causes of nocturnal enuresis:
  - Urinary tract infection.
  - Fecal retention severe enough to reduce bladder volume and cause bladder neck dysfunction.
  - Polyuria from osmotic diuresis (DM) or renal concentrating disorders (chronic renal failure).

# Subject4: Cardiorespiratory history

**1-Cough:**

- Onset (sudden, gradual).
- Duration:
  - < 2 weeks ➔ acute (bronchiolitis, pneumonia, asthma)
  - 2 weeks – 2 months ➔ acute prolonged cough (pertussis, chronic sinusitis)
  - > 2 months ➔ chronic (foreign body, GERD, Tracheo-esophageal fistula, cystic fibrosis, bronchiectasis, mucociliary dyskinesia).
- Timing:
  - Nocturnal ➔ allergy, asthma.
  - Midnight ➔ sinusitis.
  - Early morning ➔ Bronchiectasis, COPD, chronic bronchitis, adenoid, smoking.
  - Exercise ➔ asthma.
  - All over the day ➔ allergy, foreign body.
- Continuous or intractable.
- Character:
  - Barking ➔ Group, LaryngeoTracheoBronchitis.
  - Staccato ➔ Chlamydia pneumonia and respiratory infections and allergies.
  - Wheezing ➔ Bronchiolitis, asthma.
  - Paroxysmal ➔ Whooping cough, bronchiectasis, or a lung injury.
  - Paroxysmal post-tussive vomiting ➔ Pertussis.
- Wet or dry?
  - If productive (with sputum) ask about: color, amount, consistency, contains blood or clot.
  - Wet cough ➔ lower respiratory tract problem.
  - Dry cough ➔ upper respiratory tract problem.
- Important points:
  - Foreign body aspiration: for children 6 months to 6 years.
  - Recent infection, allergy, atopic family.
- Associated symptoms:
  - Runny nose, sore throat, fever ➔ nonspecific disease.
  - Headache, itchy eyes, sore throat ➔ postnasal drip.
  - Wheezing and cough with exertion ➔ asthma.
  - Night sweats ➔ TB.
  - Spitting up, irritability, or arching of the back after feedings in infants ➔ GERD.
  - Feeding problem ➔ may suggest Tracheo-esophageal fistula.
  - Abdominal pain ➔ bacterial pneumonia.
  - Weight loss or poor weight gain and foul-smelling stools ➔cystic fibrosis.
  - Muscle soreness ➔ viral illness or atypical pneumonia.

***Notes:***

- Cough caused by stimulation of nerve receptors in the pharynx, larynx, trachea or large bronchi.
- Cough center located in medulla oblongata.
- For most children, episodes of cough are due to upper respiratory tract infections caused by the common cold viruses and do not indicate the presence of a long-term or serious underlying respiratory disease.
- Causes of recurrent or persistent cough:
  - Recurrent respiratory infections.
  - Post-specific respiratory infections (pertussis, RSV, Mycoplasma).
  - Asthma.
  - Suppurative lung diseases (cystic fibrosis, ciliary dyskinesia or immune deficiency).
  - Recurrent aspiration (± gastro-esophageal reflux).
  - Persistent endobronchial infection.
  - Inhaled foreign body.
  - Cigarette smoking (active or passive).
  - Tuberculosis.
  - Habit cough.
  - Airway anomalies (tracheo-bronchomalacia, tracheo-esophageal fistula).
- Treatment of cough:
  - Management of the underlying disorder.
  - Antibiotics for bacterial pneumonia.
  - Bronchodilators and anti-inflammatory drugs for asthma.
  - Supportive care (oxygen, bronchodilators as needed) for viral infections.
  - Little evidence exists to support the use of cough suppressants and mucolytic agents.

## 2-Shortness of breath:

- Onset.
- Duration.
- Day/night.
- Degree ➔ interfere with feeding, activity, sleeping.
- Aggravating and relieving factors.
- Associated symptoms ➔ bluish discoloration, fever, sweating, abdominal pain.
- First time or have similar attack previously.
- Known case of cardiac or respiratory disease.

***Common causes of SOB in children:***

- Choanal atresia.
- Foreign body in upper airway.
- Enlarged tonsils.
- Asthma.

- Respiratory tract infection.
- Obesity.
- Congenital cyanotic heart disease.

## 3-Noisy breathing:

- Onset, duration, frequency.
- Continuous or not.
- Character of sound.
- Day or night.
- Related to allergy.
- Relation to feeding and sleeping.
- Any deformity in nose or respiratory tract.

***Site of sound:***

- Nose ➔ snoring "inspiratory", adenoid hypertrophy.
- Epiglottis ➔ Grunting "expiratory", sign of respiratory distress
- Grunting:
  - Pulmonary:
    - Effusion, pneumothorax.
    - Dramatic response to O2 for 10 minutes.
  - Extra-pulmonary:
    - Cardiac: acute heart failure.
    - Metabolic acidosis: gastroenteritis, diarrhea, diabetic ketoacidosis, salicylate.
    - Severe blood loss: anemia.
  - Expiratory phase due to respiratory failure.
- Larynx ➔ Strider "inspiratory", e.g. croup.
- Trachea and major bronchus ➔ Crepitation (by stethoscope), Rattling (by ear). "Inspiratory and expiratory" causes:
  - Bronchiolitis: viral (RSV), less than 1 year, peak at 6 months.
  - Pneumonia: more than 1 year, inflammation of lower respiratory tract.
- Minor bronchioles ➔ wheezing "mainly expiratory", causes: bronchiolitis, asthma, heart failure, pulmonary edema.

***Watch these videos:***

- https://www.youtube.com/watch?v=_kGUGgPn6_E
- https://www.youtube.com/watch?v=Qbn1Zw5CTbA&list=PLrL1lM0-SHvL2dYx-Pr7veEshYo92f3uj

## 4-Chest pain:

- Quality (sharp, stabbing, aching)
- Intensity.
- Location (can they point to location with one finger?)
- Radiation.
- Timing of the pain with relationships to exercise, eating.
- Duration, frequency.
- Ask about any history of trauma.
- What makes it better or worse?
- What do they think is causing the pain?
- Any recent stressors at home or school or problems with anxiety?

***Common causes of chest pain in children:***

- Cardiac (CAD, valve disease, structural disease).
- Musculoskeletal (fractured rib, muscle strain).
- Respiratory (asthma, foreign body in airways, chronic cough, pleural effusion).
- GIT (esophagitis, GERD).
- Non-organic (anxiety, stress).

## 5-Bluish discoloration:

- Duration.
- Site (skin, mucus membrane).
- Central or peripheral (lips, tip of tongue, peripheries).
- Respiratory or cardiac diseases.
- Any congenital anomalies.
- Associated symptoms (cough, SOB).
- Any drug use.
- Exposure to cold.
- Previous same condition.

***Notes:***

- Respiratory causes: T.B, respiratory distress syndrome, asthma, pneumonia, pneumothorax, bronchiectasis, lung abscess, cystic fibrosis, asbestosis, familial (10%).
- Cardiac disorders: cyanotic congenital heart disease.
- Acute cyanosis: pneumonia (respond to O2).
- Chronic cyanosis: Tetralogy of Fallot (not respond to O2).
- Signs of chronic cyanosis: finger clubbing, polycythemia (due to chronic hypoxia), gingival hyperplasia.
- Acrocyanosis: occur in normal baby.

## 6-Palpitation:

- Onset, duration, frequency.
- Character (told mother to tap it on the table).
- Relation to feeding or activity.
- Presence of fever.
- Any cardiac problem.
- Use of any drug.

## Notes:

- Relation of other systems to respiratory system:
  - GIT ➔ chest infection + frequent bowel motion = adenovirus.
  - GUS ➔ chest infection + UTI = adenovirus.
  - CNS ➔ chest infection + meningitis = adenovirus / Fit due to hypoxia (anxious or lethargic).
  - Skin ➔ respiratory infection + rash = meals or as complication of poliovirus vaccine.
- The most important problem that affect the neonate and that 100% ends with asthma is ➔ Broncho-pulmonary dysplasia
- In family history ask about ➔ allergic conjunctivitis, allergic rhinitis, asthma, atopic dermatitis (eczema).
- Triggers of respiratory symptoms ➔ type of heating, animal contact, ventilation, type of house cleaning.
- Feeding could lead to ➔ aspiration pneumonia, cow milk protein allergy, Tracheo-esophageal fistula.

# Subject5: CNS history

## 1-Headache:

- Headache onset, duration, and severity.
- Site (unilateral or bilateral).
- Aggravating and reliving factors.
- Associated symptoms (fever, vomiting, visual problems).
- Family history of migraines.
- Medication history.
- History of trauma.
- Factors that may have precipitated the headache (most often migraine).

***Notes:***

- Causes of headache:
  - Migraine headache.
  - Tension headache.
  - Cluster headache.
  - Sinus headache.
  - Head trauma-related headache.
  - Intracranial mass-related headache.
  - Benign intracranial hypertension.
  - Meningeal irritation.
  - Medication-overuse headache.
- In migraine attack ask about:
  - Onset and timing.
  - Number of attacks per month.
  - Aggravating and reliving factors.
  - Medications.
  - Eating contraindicated substances.
- Treatment of headache:
  - Elimination of identified precipitants
  - Lifestyle changes
  - Stress relief
  - Drugs: analgesics, triptans, isometheptene, ergotamines.

## 2-Abnormal body movements:

- Number of movements excessive or diminished?
- Age of onset:
  - Full term neonate ➔ jitteriness.
  - Infant ➔ myoclonus, athetosis, transient dystonia.
  - Older child ➔ chorea.

- Sex:
  - Female ➔ Sydenham's chorea, thyrotoxicosis.
  - Male ➔tics, tremors.
- Onset, duration
  - Acute ➔ Infection, trauma.
  - Slowly progressive.
- Change over time.
- Type of movement:
  - Rapid jerky ➔ chorea.
  - Slow movement ➔ athetosis.
  - Sustained ➔ dystonia.
- Involvement of body parts:
  - Distal limb ➔ athetosis.
  - All body parts ➔ chorea.
  - Hand ➔ focal dystonia.
- Presence of movements in sleep:
  - Seizure disorder.
  - Nocturnal myoclonus.
- Associated symptoms (loss of consciousness, cyanosis).
- Aggravating factors (environmental stimuli or emotional states).
- Reliving factors (drug, special position, suppressed voluntarily).
- Differentiate it from: breath holding attack and pseudo-epilepsy.
- Family history of a similar or related condition.

## 3-Convulsion:

- Precipitation.
- Description.
- Duration.
- Number of attacks.
- Associated with tongue bit and incontinence.
- Stoppage.
- Post-ictal state.
- Presence of fever or infection.
- Previous episodes.
- Family history.
- Head trauma, CNS disease.
- Developmental millstone.

## 4-Loss of consciousness:

- Duration, frequency, aggravating factor, body color (pale), abnormal movements, incontinence, take drugs, CVS disease, CNS disease.
- Any fit + loss of consciousness = generalized epilepsy until proven otherwise.
- Focal abnormal movement + loss of consciousness = partial epilepsy.

**5-Other symptoms:**

- Visual or hearing problems.
- Diplopia ➔ due to increased intracranial pressure for any cause.
- Numbness ➔ Unilateral, Bilateral, Symmetrical, Asymmetrical.
- Hemiplegia, Monoplegia, Tetraplegia.

**See this video:** https://www.youtube.com/watch?v=iyJ5i56HpTc